Mismatch I

Physical health

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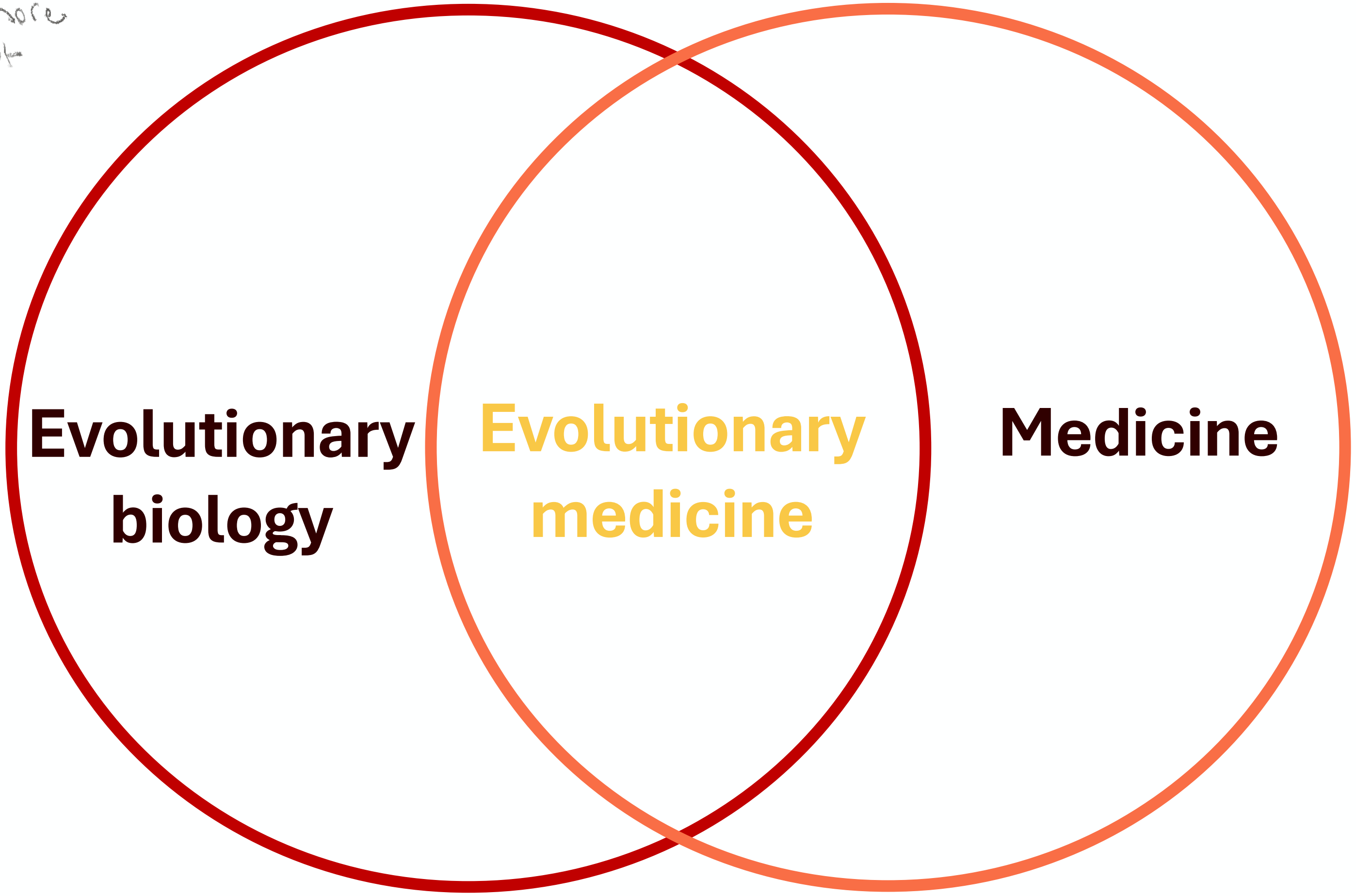
ANT215


Why is this happening?
[Insert **proximate** reason here]
Surprise
I bet you thought you'd seen the last of me
Niko Tinbergen

Evolutionary medicine uses principles of evolutionary biology to understand, treat, and prevent disease – usually from an ultimate perspective

Inform research into new treatments

Not a subfield of medicine, more of an abstract framework



Summary of Principles of the field

1. All traits need both proximate and evolutionary explanations
2. A full explanation for any trait requires answers to all four of Tinbergen's Questions
3. Traits that leave bodies vulnerable to disease have several possible kinds of evolutionary explanations
- **3.1.** The limitations of natural selection can help to explain disease vulnerabilities
- **3.2.** Mismatch between bodies and changing environments accounts for much disease.
- **3.3.** Coevolution with pathogens explains several kinds of host vulnerability
- **3.4.** Trade-offs characterise all aspects of bodies and they explain many traits that leave bodies vulnerable to disease
- **3.5.** Natural selection maximises allele transmission at the expense of health
- **3.6.** Defences provide protection in the face of threats and damage, but at considerable costs

4. Selection shapes mechanisms that mediate plasticity in various time frames
- **4.1.** Developmental Origins of Health and Disease (DOHaD) is an important cause of disease vulnerability
- **4.2.** Selection has shaped fast and slow life histories with implications for health

5. Natural selection works mainly at the level of the gene
- **5.1.** Group selection is a viable explanation only under constrained circumstances
- **5.2.** A multigenerational perspective is important

6. Kin selection can explain some traits that reduce individual reproductive success
- **6.1.** Natural selection continues to act after menopause
- **6.2.** Weaning conflicts are inevitable
- **6.3.** Conflicts between maternal and paternal genomes can cause disease

7. Control of cell replication is crucial for metazoan life
8. Intragenomic conflicts can influence health
9. ... genotypes during the lifetime of an individual
10. Natural selection shapes life history traits
11. ... effects can be selected for if they offer compensating benefits
12. Cliff-edged fitness landscapes can account for the persistence of some genetic diseases
13. Attention to ethics is important
14. Races are not biological categories
15. Genetic differences between human subgroups influence health
16. It is a mistake to assume that what is is what ought to be
17. Natural selection is not over for humans
18. ... tracing relationships and phylogenies have many applications in evolutionary medicine
- **18.1.** ... medically relevant
- **18.2.** Phylogenetic methods can trace the origins and spread of pathogens

19. ... and testing evolutionary hypotheses remain under development
20. Organic complexity is different in kind from the complexity in machines

1. Selection operates to enhance **fitness**, not primarily to enhance health or longevity

2. Our (1) evolutionary history, and our (2) personal history through our own life cycle, does not *cause* disease (with the exception of some single gene defects), but influences our **susceptibility to ill health in particular environments**

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Why susceptibility?

1	Mismatch
2	Excessive defence mechanisms: inappropriate deployment of processes that evolved as an adaptation
3	Co-evolutionary considerations: losing the evolutionary arms race against microbes
4	Constraints imposed by our evolutionary history
5	Sexual selection and its consequences
6	Balancing selection maintaining an allele that raises disease risk
7	Demographic history and its outcomes

Tend to be homeostatic

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Smoke detector principle:

Cost of running: 200 calories
Cost of not running: 200,000 calories

a rare occurrence could kill you very easily so it makes sense to have a low threshold for setting in defense mechanism

of death

Because of the risk, system is set to a threshold that yields many false alarms

Fever, nausea, vomiting, diarrhea, anxiety,...

You can safely dampen responses given overreaction

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Pathogens evolve many times faster than humans

- Generation time of *E coli:* 15-20 minutes
- Generation time of humans: 22-33 years
  - Pathogens evolve ways around our defences

The arms race between defences and counter-defences results in dangerous mechanisms.

- Smoke-detector principle
- Autoimmune diseases

Intense responses make sense in low chance that microbes evolve to be super dangerous

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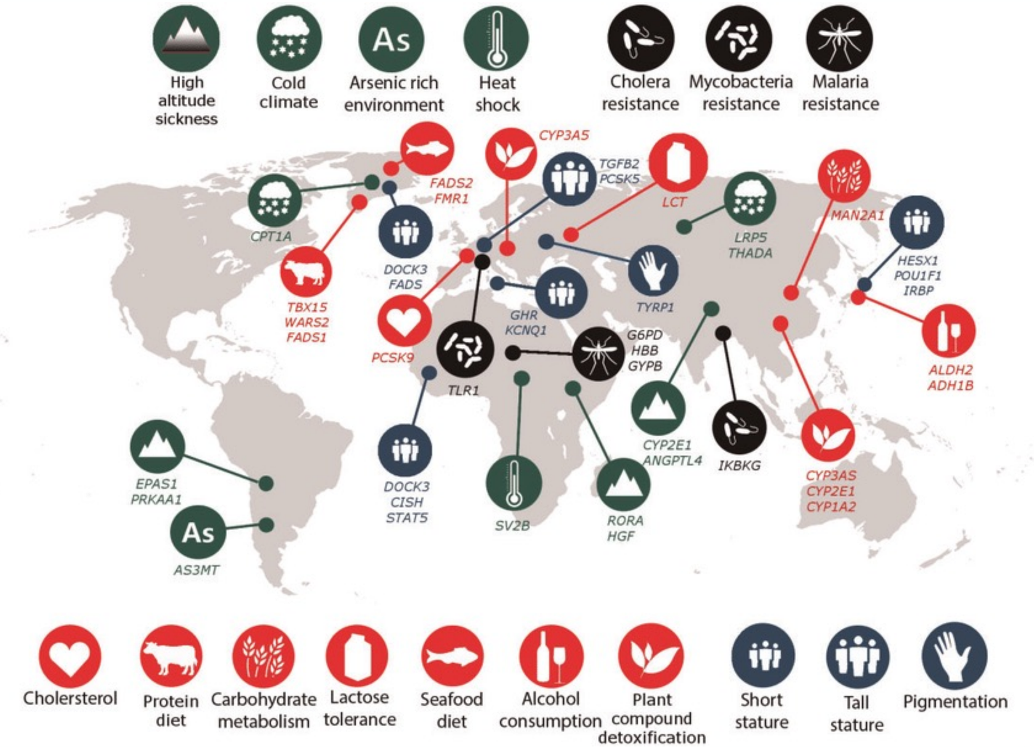
Review: Lecture 8



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Human feet suck due to primate feet being like hands but to be bipedal we've had to evolve tight ligaments and arches but this is not efficient

5 Sexual selection and its consequences

6 Balancing selection maintaining an allele that raises disease risk

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Evolutionary medicine
Evolution ward
Discussion

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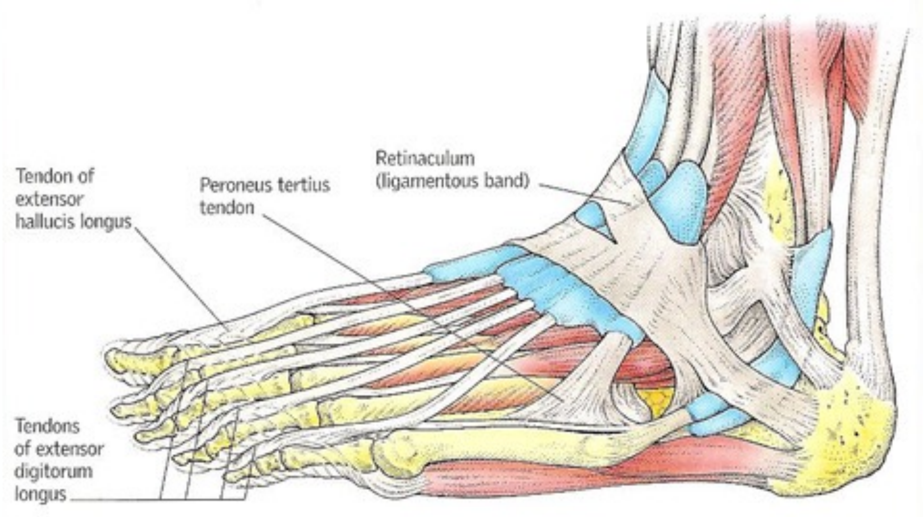
Ancestral ape condition:
Mobile grasping foot, highly mobile, moving parts

- Sprained ankles
- Heel spurs
- Collapsed arches
- Achilles tendonitis



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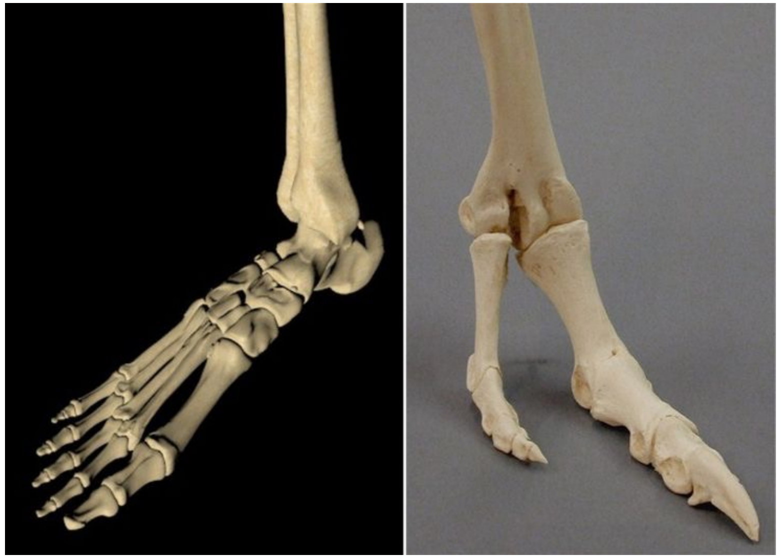

Δ ostrich feet fewer moving parts on ligaments what can fail

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7 Demographic history and its outcomes

Traits that increase reproductive output may decrease health or longevity

Review: Lecture 20

Men at all ages have higher mortality than women

Testosterone benefits:

- Investment in competition -> fitness pay-offs

Testosterone costs:

- Risk taking: accidental death
- Immunosuppressant: susceptible to infectious disease

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For heterozygosity

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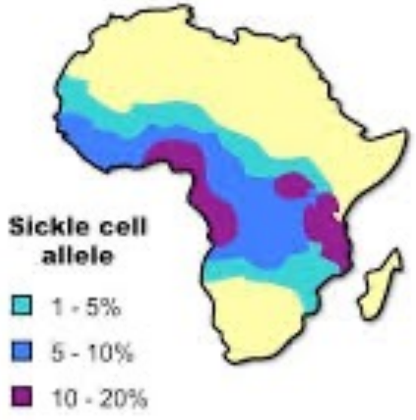

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Mismatch: two types

Evolutionary mismatch

Mismatch between evolutionary environment and current environment

Developmental mismatch

Mismatch between developmental environment and current environment

Adaptive processes lead to issues

mis match
↳ maladaptation

Developmental plasticity → adaptation
but [illegible] [illegible]
[illegible] not adapted anymore

Evolutionary mismatch

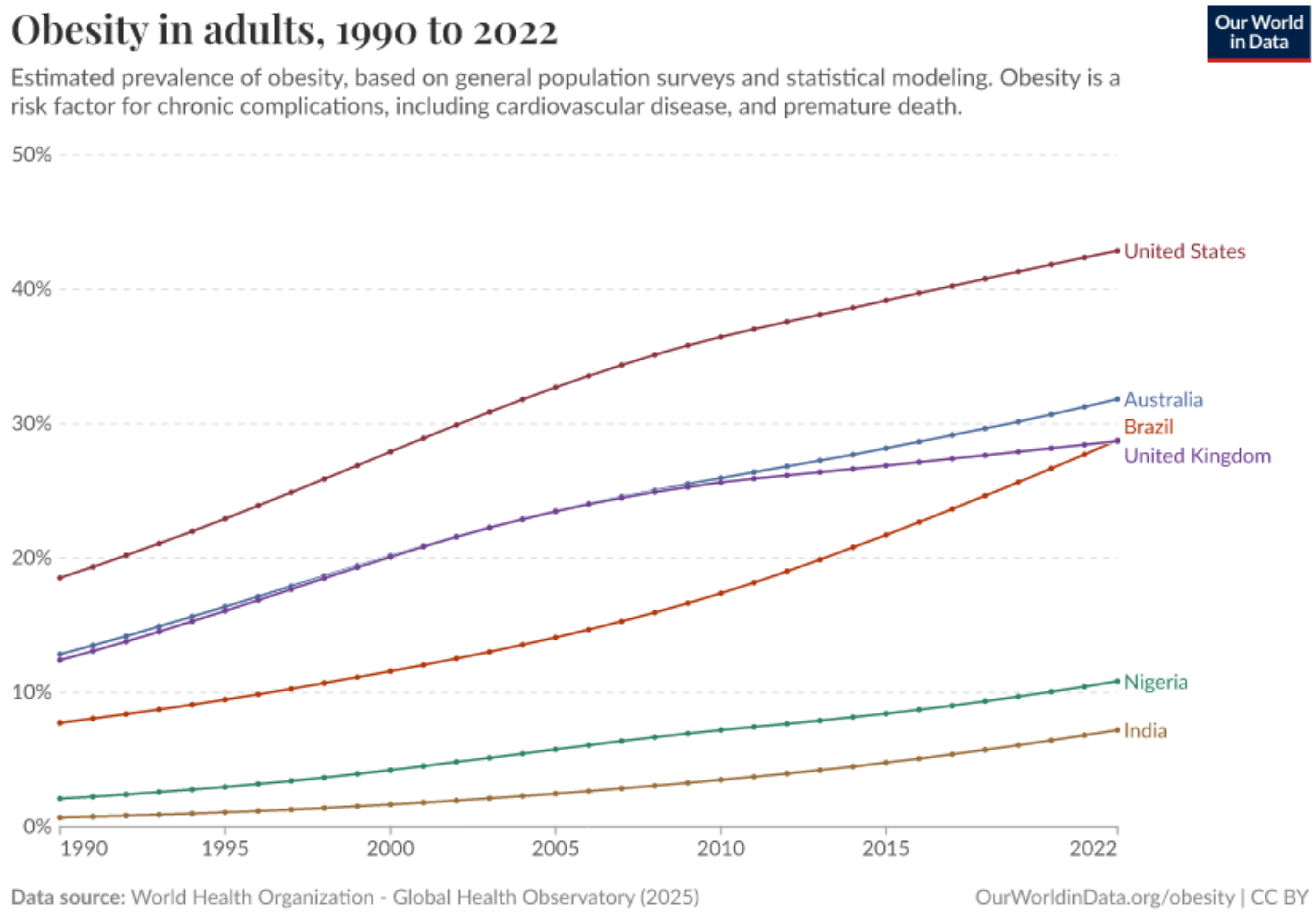
Mismatch between evolutionary environment and current environment

EXIT 247
EXIT 246
NORTH
75
85
Chattanooga
Greenville
20
Birmingham
Augusta
1/2 MILE
Fulton St
Central Ave
Ga State Univ
EXIT ONLY
LEFT
HOV EXIT
Memorial
Drive
Central Ave
Downtown/GWCC
Mercedes-Benz Stadium


Interested?
Profs Colin Shaw (Zurich) & Danny
Longman (Loughbourough)

Evolutionary mismatch

Mismatch between evolutionary environment and current environment

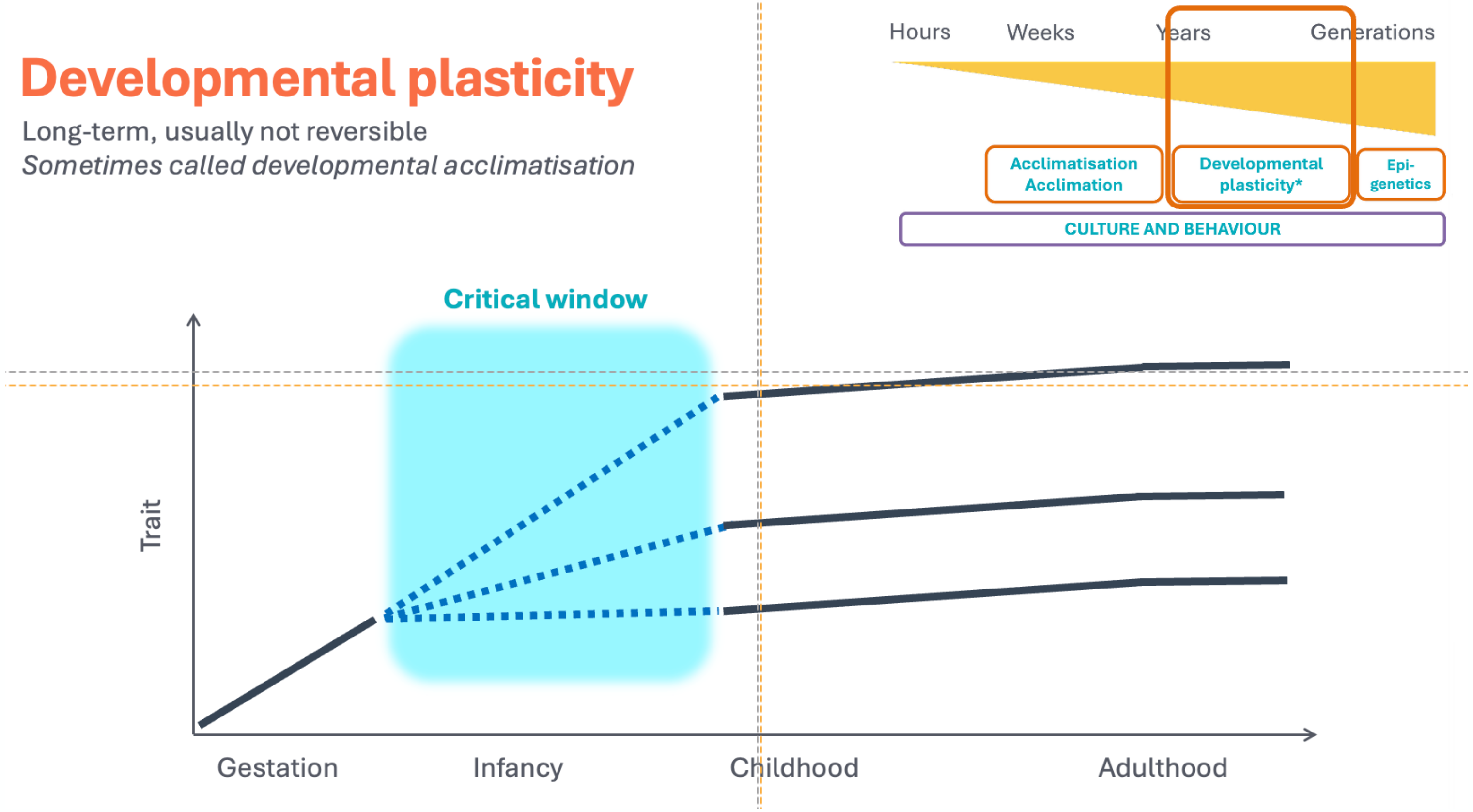


Not evolved for processed food

↳ partially true but we do not know what we ate in environment where we adapted

Developmental mismatch

Mismatch between developmental environment and current environment



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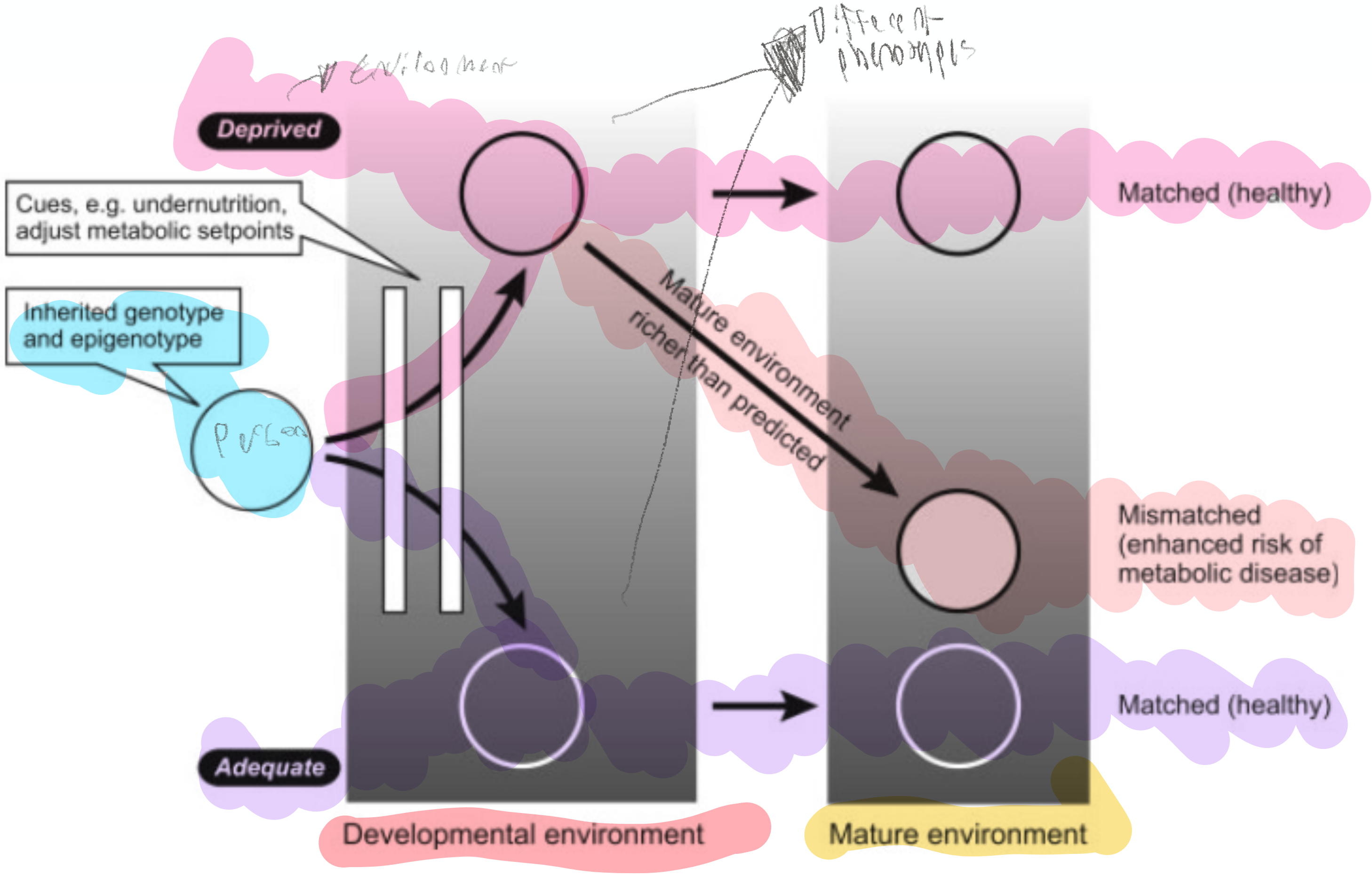


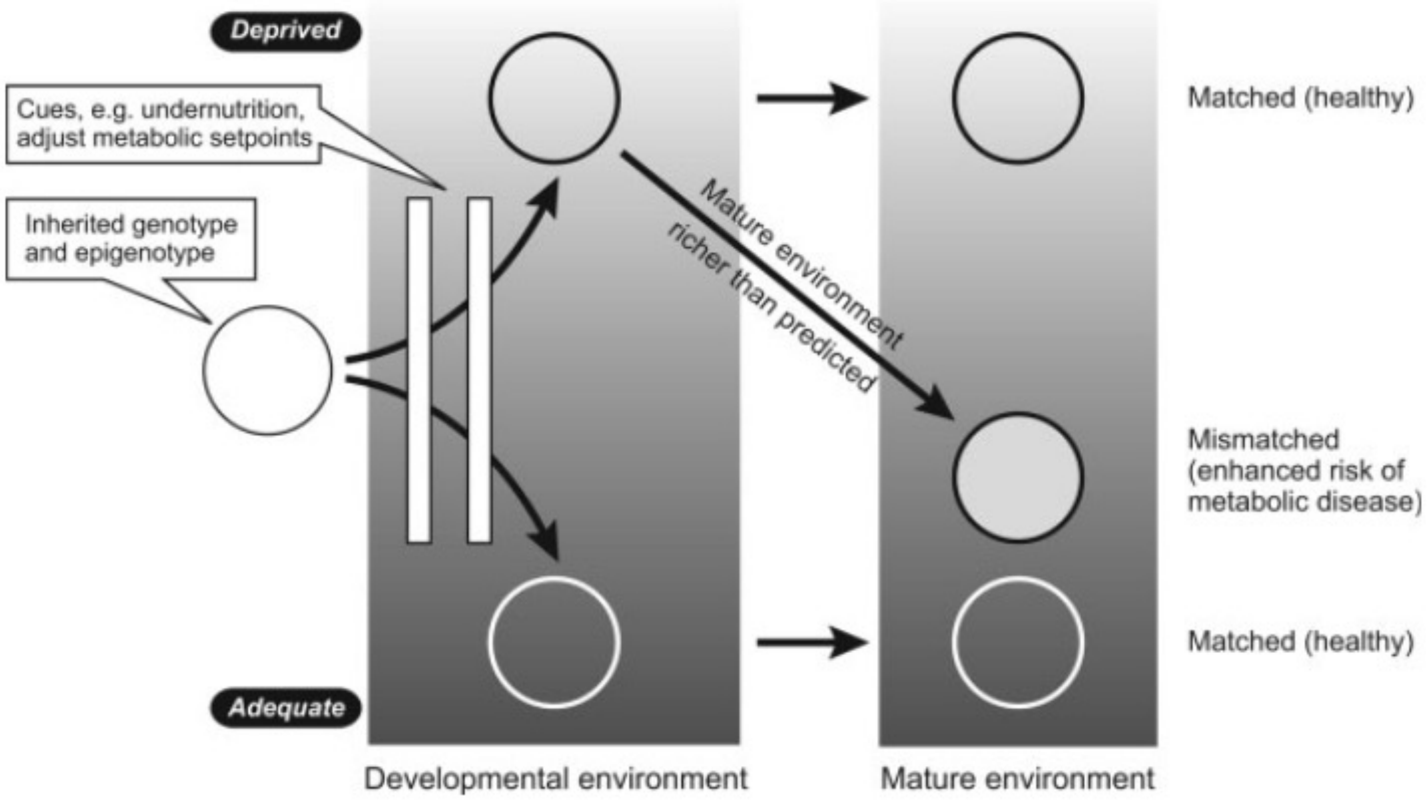
DOHaD paradigm

Developmental Origins of Health and Disease

Gluckman PD, Hanson MA, Beedle AS. Early life events and their consequences for later disease: a life history and evolutionary perspective. Am J Hum Biol. 2007;19(1):1–19.

Developmental mismatch

Mismatch between developmental environment and current environment



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Evolution ward

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Mismatch between evolutionary environment and current environment

Developmental mismatch

Mismatch between developmental environment and current environment

- 3 case studies; 10 minutes to discuss each.
- Your group presents their last case to the whole class – followed by a brief discussion.

Questions:

Which type of mismatch is implicated?

What public health *and* clinical medicine interventions could arise from this perspective?

What ethical concerns are raised by this perspective?

Case 2:

Miss Anna Mensis is very worried about her son, Afarensis (goes by Alfie), who is scrubbing his hands with hand sanitizer. You notice a sheen on his face: clearly, he is a particularly cleanly-kept child.

With them is Alfie's exotic cousin, Bahrelgazali, who is covered in mud, having just played outside for what feels like 3.5 million years.

"Alfie's simply always wheezing and breathless," Miss Mensis cries, "and I think it's asthma. I don't know what to do. My sister's son has never coughed once, and surely they have very similar DNA."

"I know just what has happened," you say.

What is going on?

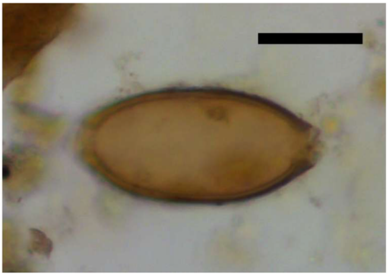
Here are some data to help you make a case:

Date	Prevalence of soil-transmitted helminth infections (global)
1990	15,930 cases per 100,000 people
2021	8,430 cases per 100,000 people

MAY 31, 2019

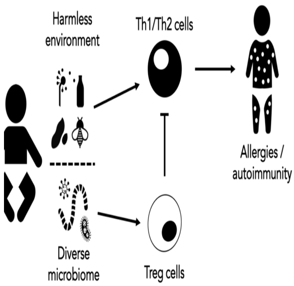
Ancient feces reveal parasites in 8,000-year-old village of Çatalhöyük

by University of Cambridge



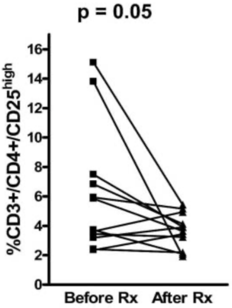
Microscopic egg of whipworm from Çatalhöyük, Turkey. Black scale bar represents 20 micrometres. Im…

New research published today in the journal *Antiquity* reveals that ancient faeces from the prehistoric village of Çatalhöyük have provided the earliest archaeological evidence for intestinal parasite infection in the mainland Near East.



Treg cells are immune cells that **inhibit the inflammatory responses that cause allergies and autoimmune disorders such as asthma**, by inhibiting the action of Th1 and Th2 cells. Treg cells are disproportionately produced during chronic exposure to helminths. For example, compare the levels of Tregs in people infected with the helminth *S. mansoni,* and the levels of Tregs in the same people after being treated for this infection:

(A)



Data from: **Watanabe, Kanji, et al. "T regulatory cell levels decrease in people infected with *Schistosoma mansoni* on effective treatment." *The American journal of tropical medicine and hygiene* 77.4 (2007): 676.**

What is a potential mismatch-based explanation for these patterns?

Hygiene, lack of exposure to Helminths causing overreactive immune response

What kind of mismatch is this case an example of?

a) Evolutionary

b) Developmental

Why?

Old friends/hygiene hypothesis, co-evolved with these Helminths which regulate immune response

How might these data be used to inform

a) Clinical practice?

Exposure to these old friends within relevant time period can try to negate overrreactive immune response

b) Public health interventions?

Generally exposing people to these microbes

Are there ethical considerations for your proposals for the use of this research in clinical practice and/or public health interventions?

You can't just give people worms and exposing people to potentially dangerous pathogens could obviously pose a potential danger

Case 3:

Your colleague is ecstatic that a child in their practice has exhibited rapid catch-up growth after their diet was supplemented with essential nutrients.

Even though maternal nutrition during the pregnancy was poor, the child is now within the normal range for weight and height.

“Great news,” you say, “but I wouldn’t celebrate just yet. Schedule that child for regular check-ups, and, in particular, keep close tabs on their insulin levels.”

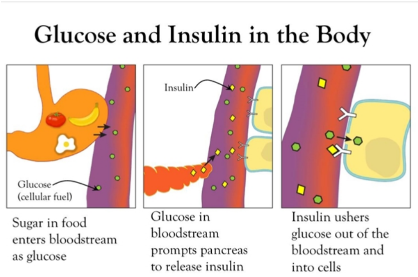
What do you know that your colleague doesn’t?

Here are some data to help you make a case:

McMillen & Robinson (2005) show that poor perinatal nutrient exposure leads to

- Decreased beta cells in the pancreas. Beta cells are cells that make insulin.
- Decreased capacity of the liver and muscles to take up glucose (insulin resistance).

McMillen, I. Caroline, and Jeffrey S. Robinson. "Developmental origins of the metabolic syndrome: prediction, plasticity, and programming." *Physiological reviews* 85.2 (2005): 571-633.



What is a potential mismatch-based explanation for these patterns?

Poor perinatal nutrient exposure but lots of nutrient exposure when older

What kind of mismatch is this case an example of?

a) Evolutionary

b) Developmental

Why?

Mismatch between perinatal nutrient exposure and current nutrient exposure

How might these data be used to inform

a) Clinical practice?

Being aware of risk factors for diabetes allowing for faster recognition and therefore faster treatment

b) Public health interventions?

Ensuring proper perinatal nutrient exposure by addressing food insecurity or inequalities

Are there ethical considerations for your proposals for the use of this research in clinical practice and/or public health interventions?

Poor perinatal nutrient exposure wouldn't be a direct determinant of diabetes so you can't treat everyone for diabetes just based on this but individuals who've experienced nutrient mismatch developmentally can be paid attention to considering whether or not they would develop diabetes to promptly address it if they do

Case 1:

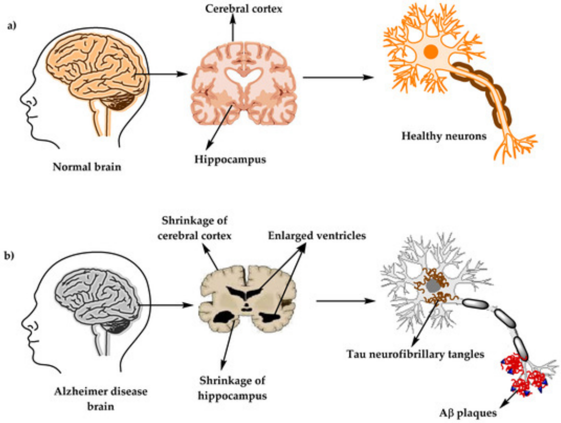
As a modern *Homo sapiens* gynaecologist, you've been around for a while: you've recently had your practice's 300,000-year anniversary.

Your practice used to be so busy that 299,950 years just flew by, but in the last half century things have substantially quietened down. With so much time on your hands you decide to start volunteering at the Alzheimer's clinic next door, which has been increasingly overloaded in the last 49 years.

"What a strange coincidence," you think, "that as soon as my gynaecology clinic started getting less busy, the neighbours' patient numbers shot up…"

What is going on?

Here are some data to help you make a case:



Alzheimer's disease (AD) is the most common neurodegenerative disorder in people today. An important proximate cause of the disease is **plaque and neurofibrillary tangle formation in the brain**: these plaques and tangles essentially impede cognitive function. In 1984, Glenner & Wong discovered that the amyloid β protein (Aβ) is the central component of these plaques and tangles. Since then, Aβ has been considered as a driver of Alzheimer's pathological processes.

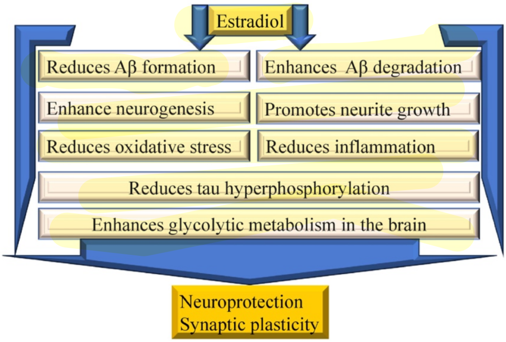


Figure from: Suganya, S., Ashok, B. S., & Ajith, T. A. (2024). A Recent Update on the Role of Estrogen and Progesterone in Alzheimer's Disease. *Cell Biochemistry and Function*, *42*(8), e70025.

Fox et al. (2013) summarise the effects of estrogens as follows:

Estrogens have myriad neuroprotective functions that defend against the AD (Alzheimer's disease) pathological cascade. Estrogens have been shown to:

- inhibit Aβ formation
- promote Aβ clearance
- inhibit neuronal apoptosis
- reduce brain oxidative stress and inflammation.

Estrogen levels during pregnancy	14.5 ng/ml
Estrogen levels during ovarian cycle	0.33 ng/ml
Months spent pregnant across the life course in Britain	21
Months spent pregnant across the life course in !Kung hunter-gatherers	56.4
Months spent pregnant across the life course in Aché hunter-gatherers	96

What is a potential mismatch-based explanation for these patterns?

Less pregnancy due to cultural factors —> less estrogen —> more Alzheimer's. Correlative but not necessarily causative

What kind of mismatch is this case an example of?

a) Evolutionary

b) Developmental

Why?

Mismatch in hormone levels due to cultural factors but hormones are beneficial

How might these data be used to inform

a) Clinical practice?

Lower estrogen levels —> higher risk for Alzheimer's which could help with early detection

b) Public health interventions?

really careful when communicating and you can't say that links should influence behavior

Increased monitoring of hormone levels and general awareness of risk factors combined with more research into Alzheimer's so with early awareness it would be beneficial

Are there ethical considerations for your proposals for the use of this research in clinical practice and/or public health interventions?

Actually counteracting this could be potentially harmful as estrogen has side effects, also increasing pregnancies would be unethical as well

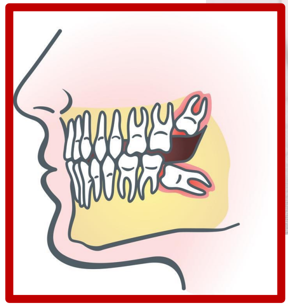
Evolution ward

Mr. Mauer comes in, complaining of pain in his mandible. You ask him what he has been eating, and he tells you he is a farmer whose favourite food is oatmeal.

His Neanderthal chaperone rolls his eyes - not that you can see it underneath his brow ridges - and chucks a tough, uncooked vegetable and some hard-shelled nuts in his chinless mouth.

You ask Mr. Mauer to open his mouth so you can examine him, and immediately notice that he only has two erupted mandibular molars, instead of three. Bingo! Wisdom teeth impaction.

What is going on?



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